

Besides the symplectic structure there are other possible choices which could underpin the design of numerical methods. For example, symplectic methods are volume preserving, but it is possible to construct volume preserving methods that are not symplectic. Similarly time-reversibility is a fundamental property of molecular dynamics models (due to the quadratic kinetic energy function) and could be used as a starting point for developing numerical schemes. We discuss these alternatives in this chapter. We also look at the performance of methods which are specifically designed to preserve the energy using a projection technique. The conclusion of this chapter is that symplectic methods are to be preferred in most cases; this observation is in agreement with current practice in molecular dynamics simulation. In molecular dynamics we typically have a restricted class of systems which are purely Hamiltonian, so geometric integration seems the natural starting point. A related concept to geometric integration is that of *variational integration* [395] which is in some ways more general (including some mechanical systems that are not in Hamiltonian form) but does not provide a useful backward error analysis (except in the same cases as geometric integration).

Geometric integration has been developed as a subject from around 1985 to the present. There remain many interesting open questions in this area. The discussion presented here is necessarily highly compressed and focussed to the issues directly relevant for molecular dynamics. For starting points to more comprehensive understanding of the topic, one may study the books [164, 227] or articles [121, 369].

3.1 The Modified Hamiltonian

We begin by asking the following question which takes us right to the heart of what is needed in molecular dynamics: how is the energy affected by numerical discretization?

Let us begin with an illustrative example. Consider the harmonic oscillator with frequency Ω which has Hamiltonian $H(q, p) = p^2/2 + \Omega^2 q^2/2$, and consider the adjoint symplectic Euler method

$$Q = q + hp, \quad P = p - h\Omega^2 Q,$$

If $H(q, p) = E$ then, for typical steps, we cannot expect $H(Q, P) = E$. (Just insert the formulas for Q and P into the Hamiltonian and check that the value is not the same as $H(q, p)$.)

However, if we modify the Hamiltonian from $H(q, p) = p^2/2 + \Omega^2 q^2/2$ to

$$\tilde{H}(q, p) = \frac{p^2 + h\Omega^2 pq + \Omega^2 q^2}{2}, \quad (3.1)$$

not be periodic and the numerical method then computes a discrete sequence of points which gradually fill in the ellipse (see the right panel of Fig. 3.1).

Obviously this conservation property (the existence of a perturbed energy surface) is a special feature of the method we have considered. If we used Euler's method to solve the harmonic oscillator we would find that energy grows without bound.

Now consider the more general Hamiltonian setting. Let $\mathcal{G}_h$ be a r th order symplectic integrator, $r \geq 1$. Suppose that it is the flow map of a certain Hamiltonian system, with Hamiltonian $\tilde{H}_h$. If the method order is r , we may expect this Hamiltonian to be a $\mathcal{O}(h^r)$ approximation of H , thus we posit an expansion of the form

$$\tilde{H}_h = H + h^r H^{(r)} + h^{r+1} H^{(r+1)} + \dots$$

To determine the terms $H^{(r)}, H^{(r+1)}, \dots$, we write the differential equations on $\tilde{H}_h$:

$$\dot{z} = J\nabla H + h^r J\nabla H^{(r)} + h^{r+1} J\nabla H^{(r+1)} + \dots$$

The solution of this system can be expanded in powers of h and equated term-by-term with the expansion of $\mathcal{G}_h$ in powers of h . In this way, successive terms may be computed. Although mechanical, this procedure is tedious. For splitting methods, a more elegant approach is available using the *Lie derivative*, which we next introduce.

3.2 Lie Derivatives and Poisson Brackets

Let $\phi(z)$ be an arbitrary smooth, scalar-valued function of the phase variables (assumed to lie in $\mathbb{R}^m$). At any point $z = \xi$ in phase space assume that there is a unique solution $z(t, \xi)$ such that $\dot{z}(t) = f(z(t))$, $z(0) = \xi$ that is globally defined for all t . Now differentiate $\hat{\phi}(t) = \phi(z(t))$ with respect to time, using the chain rule, to see how $\phi(z(t))$ is changing as t is varied:

$$\left. \frac{d}{dt} \hat{\phi} \right|_{t=0} = \left[\sum_{i=1}^m \frac{\partial \phi}{\partial z_i} \dot{z}_i \right] \bigg|_{z=\xi} = \nabla \phi(\xi) \cdot f(\xi).$$

The right hand side suggests a shorthand notation. If we define the Lie derivative $\mathcal{L}_f$ by

$$\mathcal{L}_f \phi = f \cdot \nabla \phi,$$